Policy gradient methods in reinforcement learning

Leonardo Toffalini

2026-04-16

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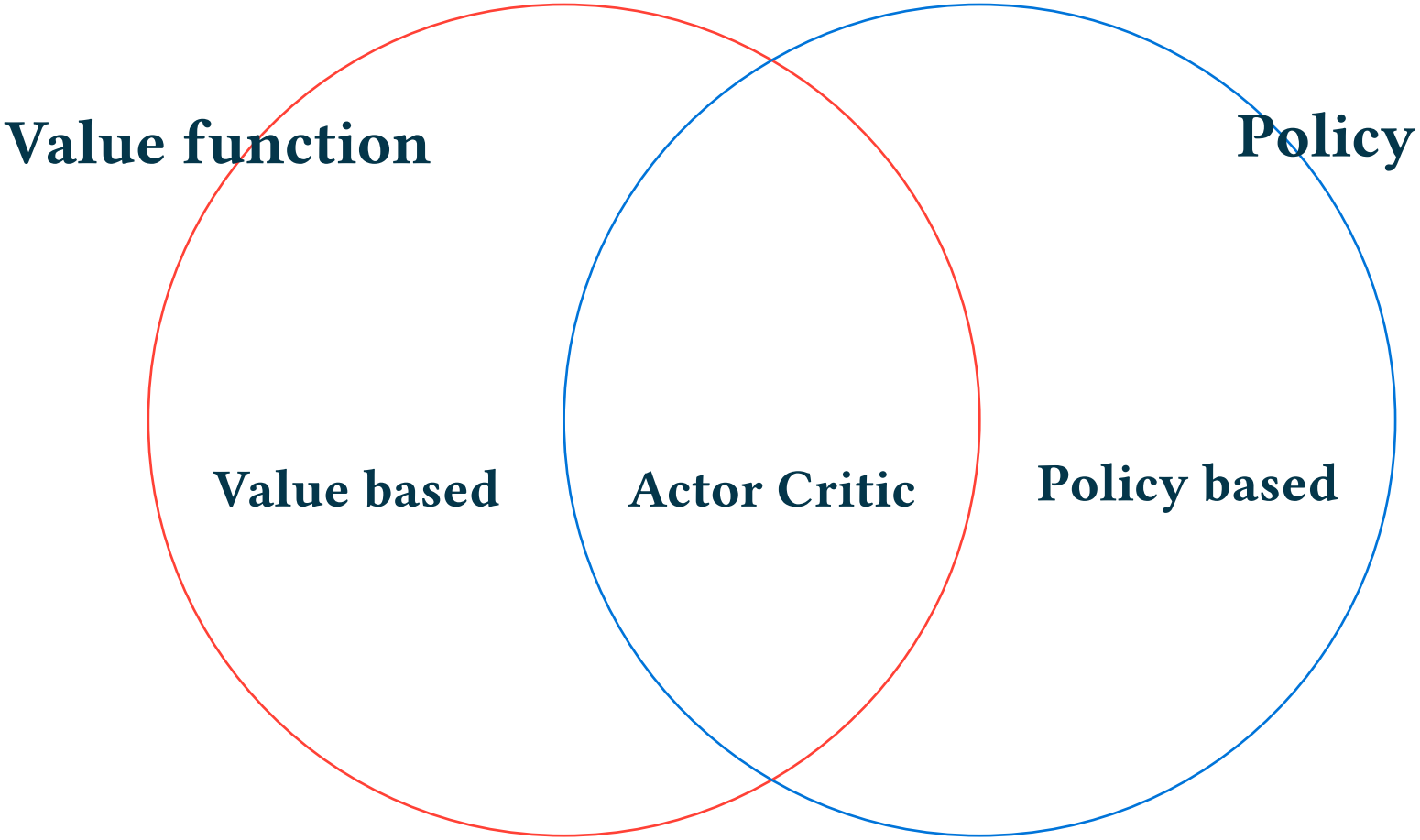
1. Policy gradient methods

1.1 Problems with value based methods

- Value iteration requires transition probabilities.
- Q-learning computes max over action space.
- What to do for huge/continuous action spaces?

1.2 Value based vs policy based methods



1.2 Value based vs policy based methods

- Value based:
  - learnt value function
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1.3 Parametric policy

- We can say that a policy depends on a set of parameters θ.
- This could be the weights and biases of a neural net.
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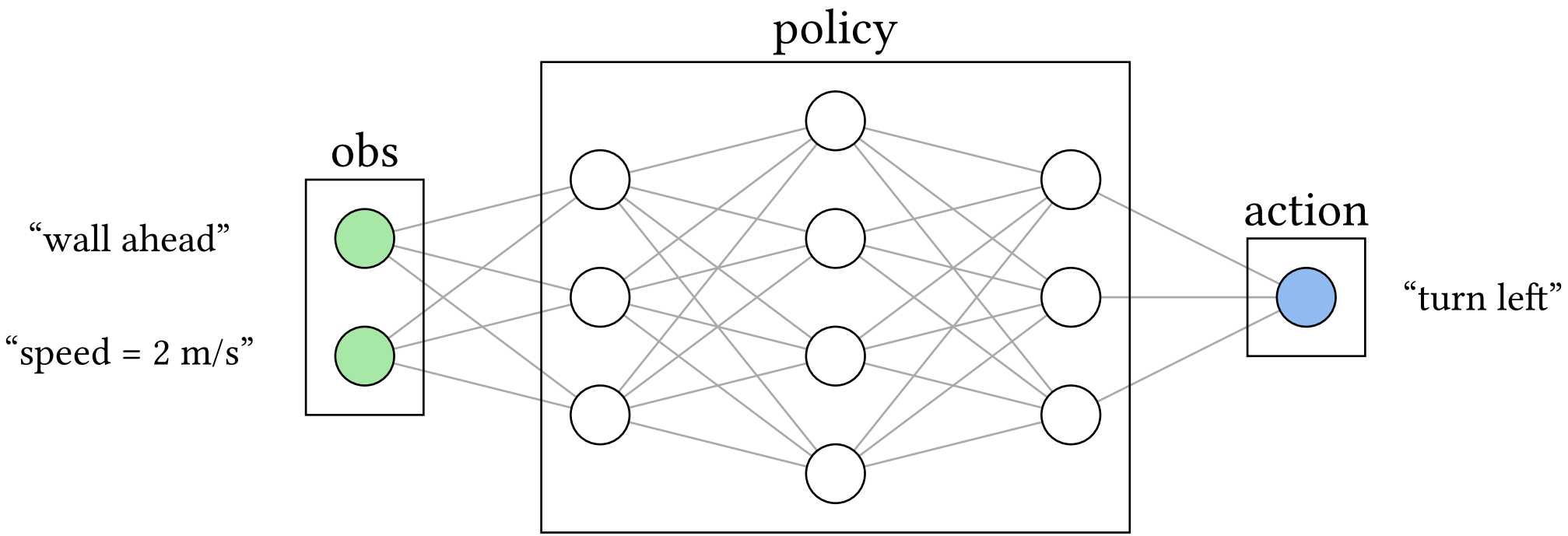

1.4 Loss functions

Define the following objective functions

$$J_1(\theta) = v_{\pi_\theta}(s_1) = \mathbb{E}_{\pi_\theta}[v_1] \qquad \text{(start value)}$$

where $d^{\pi_\theta}(s)$ is the stationary distribution of the Markov chain for π_θ.

Goal: Find θ that maximizes $J(\theta)$.

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$$J_3(\theta) = \sum_{s \in \mathcal{S}} d^{\pi_\theta}(s) \sum_{a \in \mathcal{A}} \pi_\theta(s, a) \mathcal{R}_s^a \qquad \text{(avg reward)}$$

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1.5 Policy gradient theorem

Theorem 1.5.1

$$\nabla_\theta J(\theta) \propto \sum_{s \in \mathcal{S}} d^\pi(s) \sum_{a \in \mathcal{A}} q_\pi(s,a) \nabla_\theta \pi_\theta(a|s)$$

Proof. See Sutton-Barto RL bible chapter 13.2. □

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Notice the previous theorem says that *"the gradient of the objective function is an expectation"*:

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\end{aligned}$$

1.6 Log derivative trick

With the reverse application of the chain rule we can notice the following

$$\nabla_\theta \log \pi(a|s) = \frac{\nabla_\theta \pi(a|s)}{\pi_\theta(a|s)}.$$

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Remark: Often the policy gradient theorem is stated in this form, since this is the form that is used in practice.

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Problem: In the previous formula to calculate the gradient of the objective function we need to know the action-value function q_π, instead we want to estimate it as follows

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$\Longrightarrow$ **REINFORCE**

Remark: We will implement this in the practice.

1.8 Baselines

By definition

$$q_\pi(s, a) = \mathbb{E}[G_t | S_t = s, A_t = a],$$

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Problem: What about the variance of the sample?

What if instead of $q_\pi(s,a)$ we use $q_\pi(s,a) - b(s)$ in the policy gradient theorem?

$$\nabla_\theta J(\theta) \overset{?}{\propto} \sum_{s \in \mathcal{S}} d^\pi(s) \sum_{a \in \mathcal{A}} (q_\pi(s,a) - b(s)) \nabla_\theta \pi_\theta(a|s)$$

1.8 Baselines

Split the formula

$$\sum_{s\in\mathcal{S}} d^{\pi}(s) \sum_{a\in\mathcal{A}} (q_{\pi}(s,a) - b(s))\nabla_{\theta}\pi_{\theta}(a|s) =$$

$$= \sum_{s\in\mathcal{S}} d^{\pi}(s) \sum_{a\in\mathcal{A}} q_{\pi}(s,a)\nabla_{\theta}\pi_{\theta}(a|s) - \sum_{s\in\mathcal{S}} d^{\pi}(s) \sum_{a\in\mathcal{A}} b(s)\nabla_{\theta}\pi_{\theta}(a|s).$$

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The latter part vanishes

$$\sum_{a\in\mathcal{A}} b(s)\nabla_\theta \pi_\theta(a|s) = b(s)\nabla_\theta \sum_{a\in\mathcal{A}} \pi_\theta(a|s)$$

$$= b(s)\nabla_\theta 1 = b(s)\cdot 0 = 0.$$

1.8 Baselines

With this we can see that

$$\sum_{s \in \mathcal{S}} d^{\pi}(s) \sum_{a \in \mathcal{A}} (q_{\pi}(s,a) - b(s)) \nabla_{\theta} \pi_{\theta}(a|s) =$$

$$\sum_{s \in \mathcal{S}} d^{\pi}(s) \sum_{a \in \mathcal{A}} q_{\pi}(s,a) \nabla_{\theta} \pi_{\theta}(a|s).$$

So the policy gradient theorem is true with the baseline too

$$\implies \nabla_{\theta} J(\theta) \overset{\checkmark}{\propto} \sum_{s \in \mathcal{S}} d^{\pi}(s) \sum_{a \in \mathcal{A}} (q_{\pi}(s,a) - b(s)) \nabla_{\theta} \pi_{\theta}(a|s).$$

1.9 Advantage Actor Critic

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Lets use $v_\pi(s)$ as the baseline. With it we will have $q_\pi(s,a) - v_\pi(s)$ in the policy gradient theorem, this is called the *advantage*

$$A(s,a) := q_\pi(s,a) - v_\pi(s).$$

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$$A(s,a) := q_\pi(s,a) - v_\pi(s).$$

Remark: Informally speaking, the advantage measures how much better an action a is with respect to how good the current state s is.

Problem: Do we need to estimate both q_π and v_π?
$\Longrightarrow$ Not necessarily!

1.9 Advantage Actor Critic

We can write up the definition of q_π and unroll a half step of the Bellman equation

$$\begin{aligned} q_\pi(s,a) &:= \mathbb{E}_\pi[G_t | S_t = s, A_t = a] \\ &= \mathbb{E}_\pi[R_t + \gamma v_\pi(S_{t+1}) | S_t = s, A_t = a]. \end{aligned}$$

$$\implies A(s,a) = \mathbb{E}_\pi\left[R_t + \gamma v_{\pi(S_{t+1})} | S_t = s, A_t = a\right] - v_\pi(s)$$

Let us define the *temporal difference (TD) error*

$$\delta_\pi := R_t + \gamma v_\pi(S_{t+1}) - v_\pi(S_t).$$

1.9 Advantage Actor Critic

With the TD error we get the following reformulation for the advantage

$$\mathbb{E}_\pi[\delta_\pi | S_t = s, A_t = a] = \mathbb{E}_\pi[R_t + \gamma v_\pi(S_{t+1}) - v_\pi(S_t) | S_t = s, A_t = a]$$

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The previous means that the TD error is an unbiased estimator for the advantage. With this said, we can give the formula for the gradient of the objective yet again

$$\nabla J(\theta) = \mathbb{E}_\pi[\delta_\pi \nabla \log \pi(a|s)].$$

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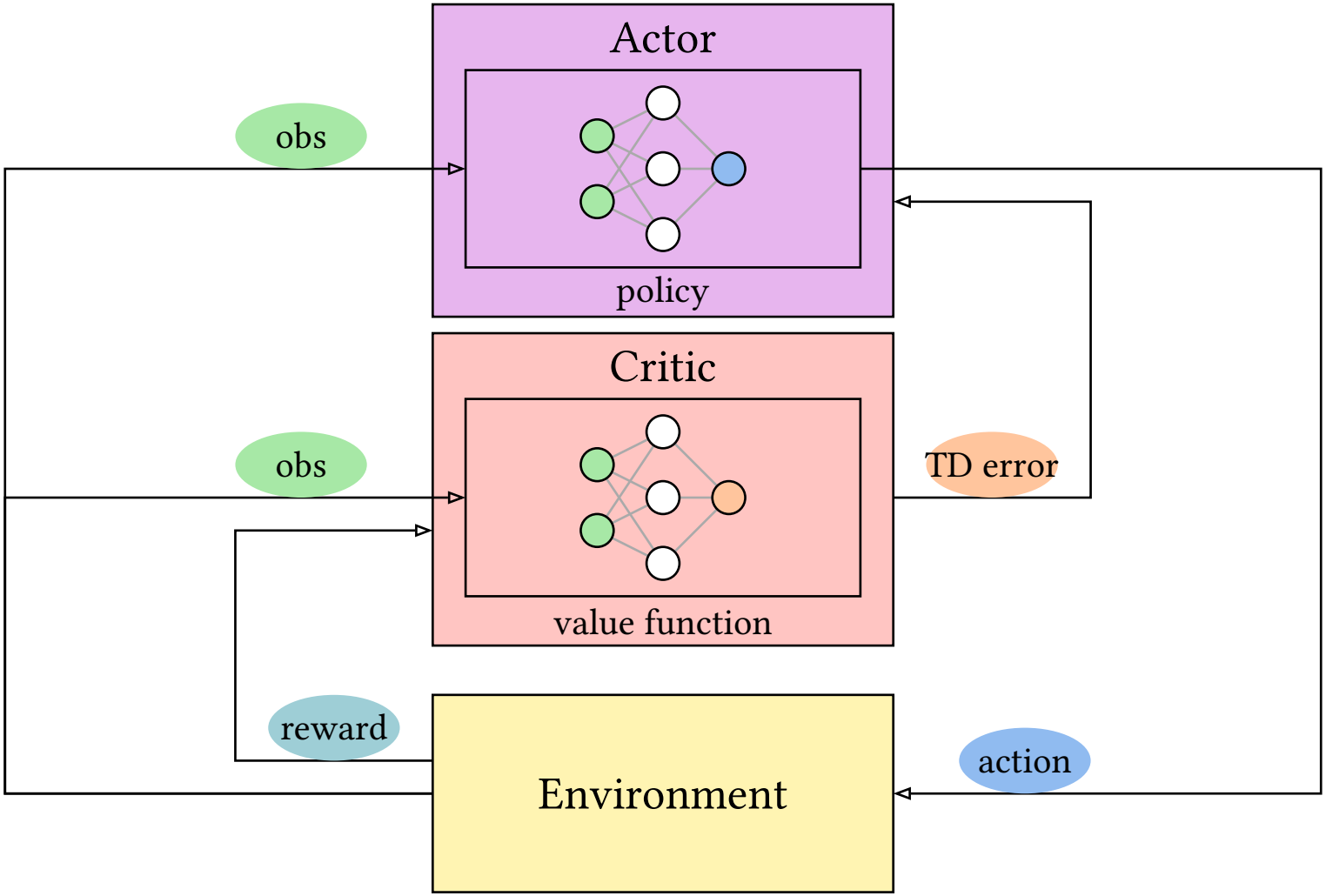


1.9 Advantage Actor Critic

Having established the reformulation of the gradient step only in terms of the state-value function it is sufficient to estimate only that.

Problem: How do we estimate the state-value function?

$\Longrightarrow$ Homework for the curious.

Problem: The TD error is an unbiased estimator for the advantage *only if* we have access to the true value function v_π, but we only have an estimate of that too: $\hat{v}_\pi$.

$\Longrightarrow$ If $\hat{v}_\pi$ has high bias then so will $\hat{A}(s, a)$.

1.10 Generalized advantage estimator

The advantage estimate had the issue that if we estimate $\hat{v}_\pi$ then the bias of that estimate will propagate to the advantage estimate $\hat{A}$.

$\Longrightarrow$ We can reduce this bias by taking more steps as follows

$$\begin{aligned}\hat{A}_t^{(k)} &:= \sum_{l=0}^{k-1} \gamma^l \delta_{t+l} \\ &= -\hat{v}_\pi(S_t) + R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \ldots + \gamma^k \hat{v}_\pi(S_{t+k}).\end{aligned}$$

We can see that the contribution of $\hat{v}_\pi$ is discounted by a factor of γ^k, thus the bias will be reduced.

Problem: If we take many steps then the variance of the estimate will increase. How do we find a balance between variance and bias?

1.10 Generalized advantage estimator

$\Longrightarrow$ Take an exponentially-weighted average of the k-step estimators:

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} := (1-\lambda)\left(\hat{A}_t^{(1)} + \lambda \hat{A}_t^{(2)} + \lambda^2 \hat{A}_t^{(3)} + \dots\right)$$

Proposition 1.10.1

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

1.10 Generalized advantage estimator

$$
\begin{aligned}
\hat{A}_t^{\gamma,\lambda} &:= (1-\lambda)\sum_{k=1}^{\infty}\lambda^{k-1}\hat{A}_t^{(k)} \\
&= (1-\lambda)\sum_{k=1}^{\infty}\lambda^{k-1}\left(\sum_{l=0}^{k-1}\gamma^l\delta_{t+l}\right) \\
&= (1-\lambda)\sum_{l=0}^{\infty}\gamma^l\delta_{t+l}\left(\sum_{k=l+1}^{\infty}\lambda^{k-1}\right) \\
&= (1-\lambda)\sum_{l=0}^{\infty}\gamma^l\delta_{t+l}\frac{\lambda^l}{1-\lambda} \\
&= \sum_{l=0}^{\infty}(\gamma\lambda)^l\delta_{t+l}
\end{aligned}
$$

1.11 Importance sampling

- In REINFORCE we collect a new batch for each training step.
  - This is terribly wasteful.
- We want to run multiple training steps on the same batch.
  - We have a batch $(s_t, a_t) \sim \pi_{\theta_{\text{old}}}$

Proposition 1.11.1

$$\mathbb{E}_{\pi_\theta}[f(s,a)] \approx \mathbb{E}_{\pi_{\theta_{\text{old}}}}\left[\frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)} f(s,a)\right]$$

1.11 Importance sampling

Proof.

$$
\begin{aligned}
\mathbb{E}_{\pi_\theta}[f(s,a)] &:= \sum_{s\in\mathcal{S}} d^{\pi_\theta}(s) \sum_{a\in\mathcal{A}} \pi_\theta(a|s) f(s,a) \\
&= \sum_{s\in\mathcal{S}} d^{\pi_{\text{old}}}(s) \cdot \frac{d^{\pi_\theta}(s)}{d^{\pi_{\text{old}}}(s)} \sum_{a\in\mathcal{A}} \pi_{\text{old}}(a|s) \cdot \frac{\pi_\theta(a|s)}{\pi_{\text{old}}(a|s)} f(s,a) \\
&= \mathbb{E}_{\pi_{\text{old}}}\left[\frac{d^{\pi_\theta}(s)}{d^{\pi_{\text{old}}}(s)} \cdot \frac{\pi_\theta(a|s)}{\pi_{\text{old}}(a|s)} \cdot f(s,a)\right] \\
&\approx \mathbb{E}_{\pi_{\text{old}}}\left[\frac{\pi_\theta(a|s)}{\pi_{\text{old}}(a|s)} \cdot f(s,a)\right]
\end{aligned}
$$

□

1.11 Importance sampling

Let $f(s,a) = \nabla_\theta \log \pi_\theta(a|s) A_t$, then

$$\begin{aligned}
\nabla_\theta J(\theta) &= \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A_t] && \text{(PG theoerm)} \\
&\approx \mathbb{E}_{\pi_{\text{old}}}\left[\frac{\pi_\theta(a_t|s_t)}{\pi_{\text{old}}(a_t|s_t)} \cdot \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A_t\right] \\
&\approx \mathbb{E}_{\pi_{\text{old}}}[r_t(\theta) \cdot \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A_t],
\end{aligned}$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\text{old}}(a_t|s_t)}.$$

1.11 Importance sampling

Notice the following

$$
\begin{aligned}
\nabla_\theta r_t(\theta) &= \nabla_\theta \left(\frac{\pi_\theta(a_t|s_t)}{\pi_{\text{old}}(a_t|s_t)} \right) \\
&= \frac{1}{\pi_{\text{old}}(a_t|s_t)} \nabla_\theta \pi_\theta(a_t|s_t) \\
&= \frac{\pi_\theta(a_t|s_t)}{\pi_{\text{old}}(a_t|s_t)} \cdot \nabla_\theta \log \pi_\theta(a_t|s_t) \\
&= r_t(\theta) \nabla_\theta \log \pi_\theta(a_t|s_t).
\end{aligned}
$$

With this we have

$$
\nabla_\theta J(\theta) \approx \mathbb{E}_{\pi_{\text{old}}}[\nabla_\theta r_t(\theta) A_t] = \nabla_\theta \mathbb{E}_{\pi_{\text{old}}}[r_t(\theta) A_t]
$$

1.11 Importance sampling

With the previous arguments we can see that the following loss function has approximately the same gradient as the original

$$L^{\mathrm{CPI}}(\theta) = \mathbb{E}_{\pi_{\mathrm{old}}}[r_t(\theta)A_t].$$

This loss function is called *conservative policy iteration.*

Since we cannot compute $\mathbb{E}[\cdot]$ exactly and we don't know A_t exactly we must approximate $\mathbb{E}$ with the sample average and A_t with $\hat{A}_t$.

Intuition: We reuse data sample with the *old policy* but we want expectation under the *new policy*. For this we correct with *how much more likely* we are to take action a at state s following the new strategy than with the old strategy.

1.12 Trust region

During training $r_t(\theta)$ becomes either huge or tiny:

- If $A_t > 0$, then optimizer moves $r_t \to \infty$.
- If $A_t < 0$, then optimizer moves $r_t \to 0$.

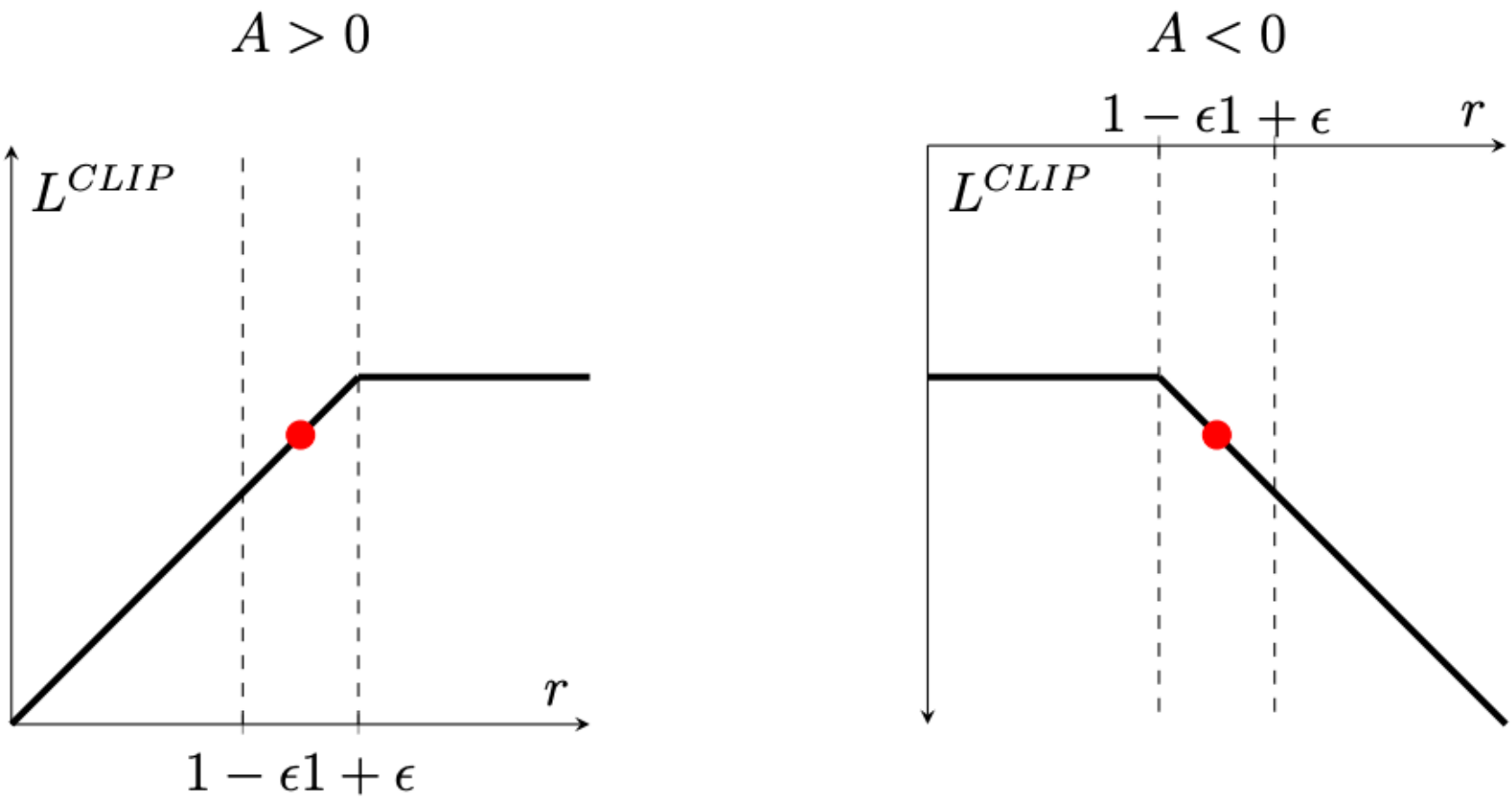
Solution: **TRPO**, **PPO**.

PPO: Let's prevent huge updates by clipping the ratio.

We define the *PPO clipped surrogate loss function* as follows

$$L^{\text{CLIP}}(\theta) = \mathbb{E}[\min(r_t(\theta)A_t, \quad \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)A_t)].$$

1.13 Clipped surrogate



1.14 Recap

- Value based methods don't work well for big or continuous action spaces.
  - Policy gradient methods.
- REINFORCE
  - Maximize returns weighted by log-probs.
- Advantage
  - Reduce variance.
- Importance sampling.
  - Reuse old data, correct with $r_t(\theta)$.
- Trust region.
  - Keep policy changes conservative.
- Clipped surrogate (PPO)
  - Simplest was to enforce trust region.